

Automated minus reference counts on the gold standard: protocol change

tranche mixed_gold_cont120_03 - frozen baseline versus candidate protocol on the same manifest and gold - misses and abstentions scored as 0 - square-root axes - every asserted gold row

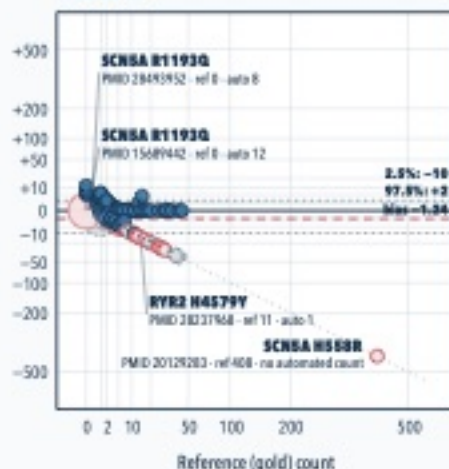
● automated count supplied ○ variant matched, no automated count (scored 0) ○ variant not found (scored 0)

Marker area - gold rows at one coordinate (wedges split it by status):

○ 1 ○ 8 ○ 28

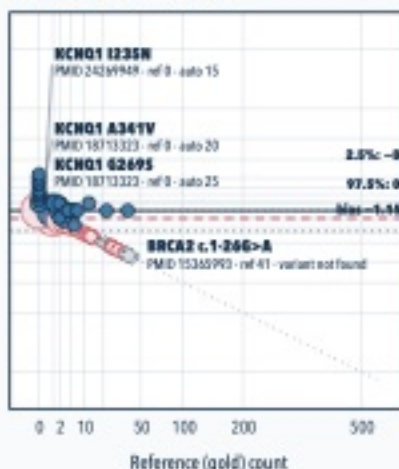
Baseline arm - 20260905_protocol_cont120_03_baseline - runtime e620f992 - locked 2026-09-06 - 120 gene-paper attempts

Carriers



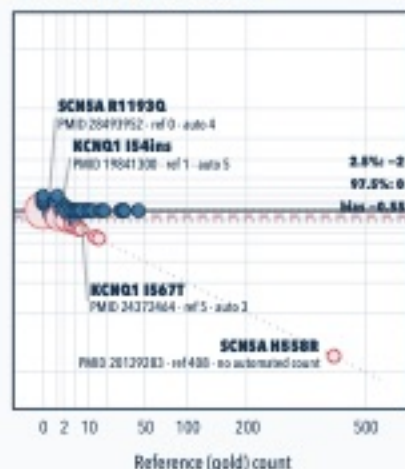
959 gold rows - matched 80.6% - supplied 60.4%
exact 67.8% (zero-baseline 14.8%) - within ± 1 88.4%
bias -1.24 - c2e MAE 1.531 - cond. MAE 0.316 (n 579)
empirical 95% limits -10 to +2 - RMSE 13.81

Affected individuals



959 gold rows - matched 80.6% - supplied 14.0%
exact 35.3% (zero-baseline 24.2%) - within ± 1 89.2%
bias -1.15 - c2e MAE 1.385 - cond. MAE 0.910 (n 134)
empirical 95% limits -8 to 0 - RMSE 4.17

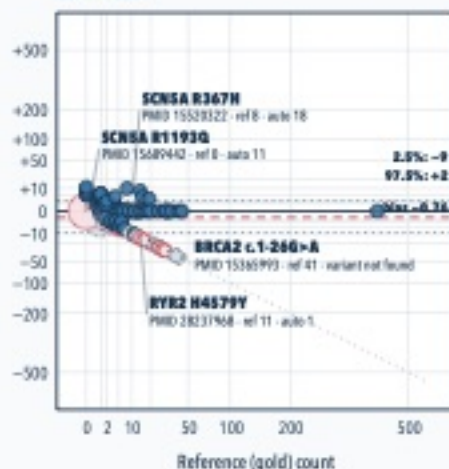
Unaffected individuals



959 gold rows - matched 80.6% - supplied 14.5%
exact 54.0% (zero-baseline 80.1%) - within ± 1 96.5%
bias -0.55 - c2e MAE 0.577 - cond. MAE 0.101 (n 139)
empirical 95% limits -2 to 0 - RMSE 13.21

Candidate arm - 20260905_protocol_cont120_03_candidate - runtime e9fef8eb - locked 2026-09-06 - 120 gene-paper attempts

Carriers



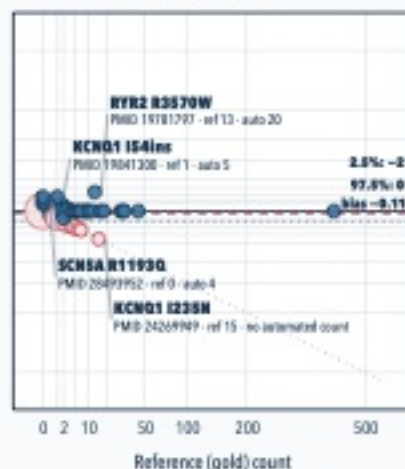
959 gold rows - matched 80.0% - supplied 60.7%
exact 67.9% (zero-baseline 14.8%) - within ± 1 88.1%
bias -0.76 - c2e MAE 1.162 - cond. MAE 0.352 (n 582)
empirical 95% limits -9 to +2 - RMSE 4.12

Affected individuals



959 gold rows - matched 80.0% - supplied 13.2%
exact 35.9% (zero-baseline 24.8%) - within ± 1 89.5%
bias -1.15 - c2e MAE 1.323 - cond. MAE 0.748 (n 127)
empirical 95% limits -6 to 0 - RMSE 3.98

Unaffected individuals



959 gold rows - matched 80.0% - supplied 14.3%
exact 54.1% (zero-baseline 80.4%) - within ± 1 96.9%
bias -0.11 - c2e MAE 0.151 - cond. MAE 0.139 (n 137)
empirical 95% limits -2 to 0 - RMSE 0.84

Candidate - baseline on the same 2877 gold rows (paired by gold row; deltas are candidate minus baseline)

Carriers

c2e MAE 1.531 $\rightarrow$ 1.102 (-0.429) - registered UB +0.041
supplied 579 $\rightarrow$ 582 - exact 650 $\rightarrow$ 651 - bias -1.24 $\rightarrow$ -0.76
rows better 11 - worse 9 - unchanged 939
coverage on matched 74.9% $\rightarrow$ 75.9% (+0.98 pp) - LB -0.35 pp

Affected individuals

c2e MAE 1.385 $\rightarrow$ 1.323 (-0.062)
supplied 134 $\rightarrow$ 127 - exact 339 $\rightarrow$ 344 - bias -1.15 $\rightarrow$ -1.15
rows better 14 - worse 9 - unchanged 936
coverage on matched 17.3% $\rightarrow$ 16.6% (-0.78 pp)

Unaffected individuals

c2e MAE 0.537 $\rightarrow$ 0.151 (-0.425)
supplied 139 $\rightarrow$ 137 - exact 901 $\rightarrow$ 902 - bias -0.55 $\rightarrow$ -0.11
rows better 5 - worse 4 - unchanged 950
coverage on matched 18.0% $\rightarrow$ 17.9% (-0.12 pp)

Identity (primary endpoint): recall 80.60% $\rightarrow$ 79.98% (-0.63 pp; LB -2.12 pp) - precision 85.32% $\rightarrow$ 85.03% (-0.29 pp; LB -2.10 pp) - registered decision: reject_or_revise_candidate - secondary counts